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FACULTY OF OF INFORMATION TECHNOLOGY

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Multi-set Ensemble Clustering for EEG/ERP

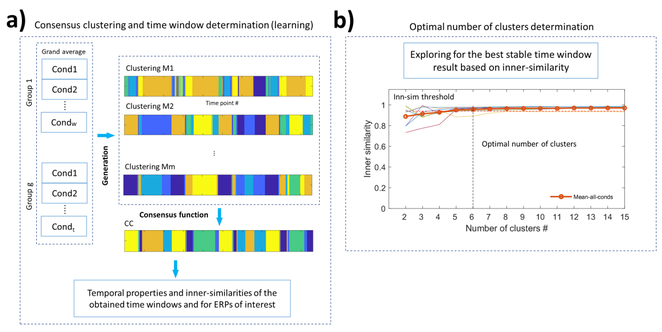
ABSTRACT

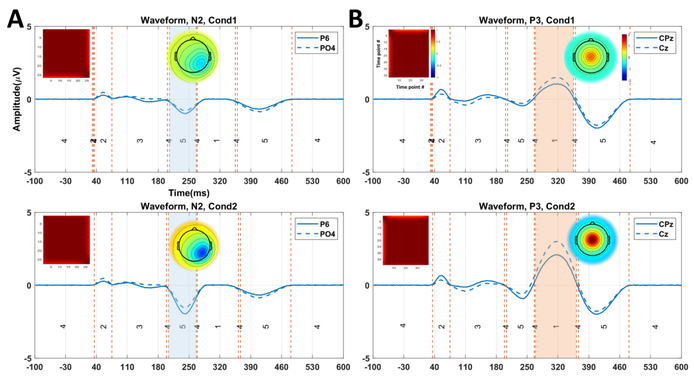
Event-related potential (ERP) techniques are a powerful tool for studying cognitive, motor, and sensory processing of brain functions. It is also well-known that all brain works, and information processing depends on brain responses at a specific time window. Here, we design a multi-subject consensus clustering from multiple-clustering results to explore reliable pieces of information shared from a set of similar sources (e.g., subjects' brains). A time window determination method has been designed using the idea of estimating stable electrode configuration in milliseconds of consecutive time bins in the perfect cluster map (i.e., cluster with high inner-similarity). Therefore, all the time points in the time window for identifying the ERP component share the same spatial (topographical) and temporal dynamics from the multiple subjects. Our findings reveal that the temporal and spatial dynamics of the brain response can be reliably explored using the proposed methodology, especially when there is little information available about measuring the cognitive process .

RESEARCH QUESTIONS AND THE PROPOSED APPROACHES

1. HOW MANY CLUSTER MAPS CAN OPTIMALLY REPRESENT THE ERP COMPONENT OF INTEREST?

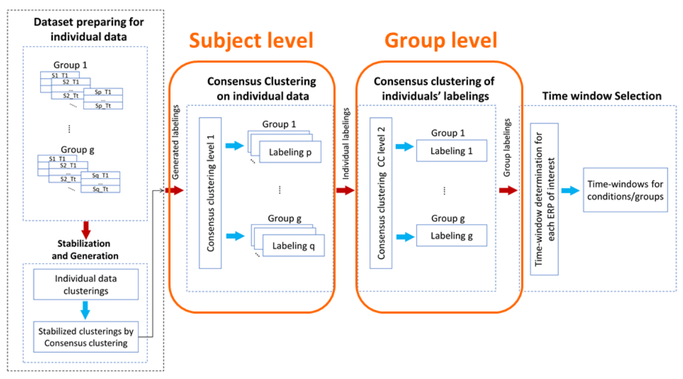
Clustering is a promising tool for determining the time window of ERP of interest. Notably, the number of clusters impacts the compression level of data following the qualifying ERP components. We design a consensus clustering to cluster Spatio-temporal ERP and estimate a proper number of clusters by assessing the quality of the determined time windows in many independent runs. The optimal number of clusters can be estimated wherein the number of clusters options (e.g., 2 to 15 clusters), the inner-similarity of the obtained time windows for different conditions/groups meet in a stable and maximum (Mahini et al. 2019). The major contributions of this study are: i) consensus clustering of ERP data; ii) time window identification for ERP of interest; iii) the optimal number of clusters determination.

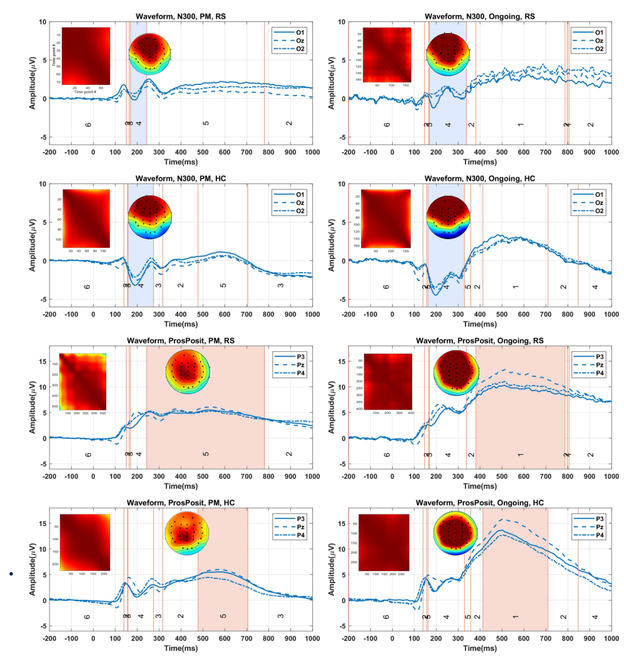




2. DETERMINATION OF THE TIME WINDOW OF EVENT-RELATED POTENTIAL USING MULTIPLE-SET CONSENSUS CLUSTERING

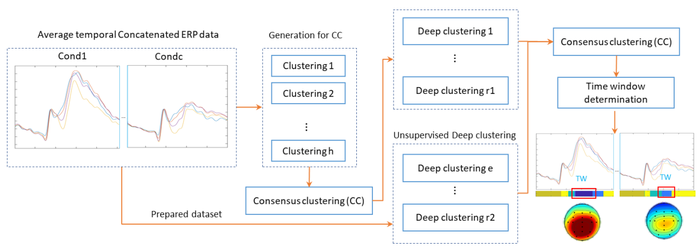
Measuring the ERP of interest merely from the grand average ERP ignores the information from the individual subjects. The ERP of interest involves the individual subjects' data, although it is not entirely identical for all subjects but similar. A multi-set consensus clustering method has been designed using ensemble clusterings in the individual subjects level and group level (Mahini et al. 2020). Therefore, consensus clustering is applied to the individual data from the subjects at the first level. Next, the obtained results are employed as an input for the consensus clustering at the second level. Thus, the proposed method explores the aggregated information between the individual subjects to determine the proper time window for ERP of interest. The proposed method has been applied to the real ERP data (Chen et al., 2015) to assess identified components.

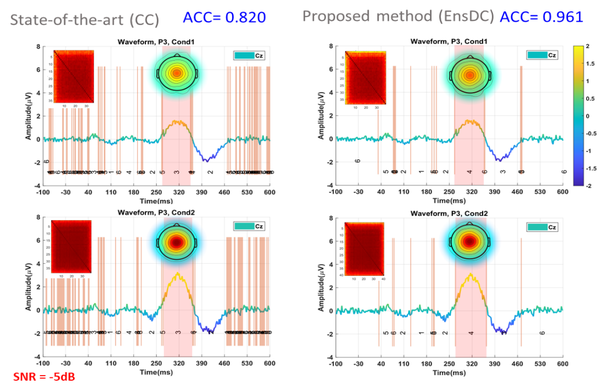




3. ENSEMBLE DEEP CLUSTERING FOR ESTIMATING TIME WINDOW OF ERP OF INTEREST

We design an ensemble cluster learning mechanism from two groups of deep clustering methods, namely supervised and unsupervised methods, to cluster Spatio-temporal ERP data. The different additional noise levels, i.e., signal-to-noise ratio (SNR) from 20, 15, 10, 5, and 0 dBs, are investigated to identify the ERP of interest from the ERP data. The yielded time window determination via the proposed method reveals more reliable isolation of P3 components in either real or noisy environments compared with the conventional state-of-the-art clustering methods. Our results revealed that the ensemble deep clustering method is a promising reliable tool for learning and extracting effective features of ERP, especially when the data quality is uncertain.





CONCLUSIONS

- → **We conclude that estimating an optimal number of clusters provides a better data compression, resulting in a proper time window for qualifying ERP components.**
- → **In addition, multi-subject consensus clustering provides a powerful tool to extract the mutual information from a set of subjects to extract similar brain responses from all the subjects.**
- → **The newly designed ensemble deep clustering is a suitable machine learning tool for learning and exploring the trained rich information about the data in the latent space for further analyses.**

Two toolboxes have been developed under the MATLAB platform, OptNC_ERP(https://zenodo.org/badge/latestdoi/197860407) and ERP_CC (https://doi.org/10.5281/zenodo.3541576) available on simulated ERP data, beside EEGLAB. Find more about us at: https://reza-mahini.github.io , Email: remahini@jyu.fi